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Evaluating the Impact of Digital Elevation Models on Urban Flood Modeling: A Comprehensive Analysis of Flood Inundation, Hazard Mapping, and Damage Estimation

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Abstract
Digital Elevation Models (DEMs) play a crucial role in flood management. This study aims to assess the effect of various global DEMs (GDEMs), including ALOS-12.5 m, ALOS-30 m, SRTM-30 m, SRTM-90 m, and NASADEM-30 m, on flood risk modeling in a densely urban area. The 1D-2D MIKE FLOOD hydraulic model was employed for the flood modeling. The process involved using a high-resolution DEM (Pleiades-1A 1 m) as the reference map (RM1), along with other GDEMs, to simulate a 50-year return period flood. The performance of GDEMs was then assessed in terms of flood inundation extent, flood hazard, and flood damage estimation, assessing their accuracy against the RM1. The study also explored the trade-offs between accuracy and efficiency by examining the effects of substituting the high-resolution map with a 5-m resolution map (Res_5 m) created through resampling. Results revealed that GDEMs tend to overestimate flood extent and underestimate depth, leading to inaccurate flood risk assessments. Among the GDEMs, NASADEM-30 and SRTM-30 outperformed others in simulating flood inundation extent but resulted in a more uniform flood depth distribution; approximately 70% of the flood extent was categorized with depths of less than 0.3 m, nearly double that of the RM1. This discrepancy led to an underestimation and overestimation of higher (H3-H6) and lower (H1) hazard levels by approximately 50%, respectively. Furthermore, GDEMs significantly overestimated flood damages, with NASADEM-30 showing a 161% overestimation compared to the RM1. Ultimately, the Res_5 m was a viable alternative for urban flood simulations as it led only to a modest 6% decrease in the flood zone area.

Keywords Urban Flooding · Digital Elevation Model · Flood Extent · Flood Damage · Flood Hazard · MIKE FLOOD

1 Introduction

Over the past few years, urban inundation has emerged as one of the most common and widespread disasters leading to substantial social, economic, and environmental consequences (Li et al. 2019). Between 2000 and 2015, floods affected 2.3 billion people,

Extended author information available on the last page of the article

constituting nearly half of all climate-related disasters, and were responsible for 20.4% of fatalities and 19.3% of economic losses from such events, cumulatively costing around US$ 70.1 billion globally (Khojeh et al. 2022). Different factors, including heavy rainfall, dense population, urbanization, reduced infiltration capacity, and industrialization have exacerbated the frequency and magnitude of floods in urban zones (Patel and Srivastava 2013; Fereshtehpour and Karamouz 2018; Pasquier et al. 2019). Understanding flood risk necessitates the critical role of flood inundation mapping, which identifies potential impact zones and assesses flood hazard severity through inundation depths. To mitigate life and property loss and enhance future flood risk management, there has been a growing focus on employing mathematical and computational tools such as hydraulic modeling (e.g., McClean et al. 2020; Karamouz and Mahani 2021; Carr et al. 2024).

The performance of hydraulic modeling for floods largely depends on several key factors: the dimensional approach of the models (1D, 2D) (Pinho et al. 2015), an accurate representation of the geometric description of the floodplains and river channels (Saksena and Merwade 2015; Azizian and Brocca 2020), model assumptions, and boundary conditions (Oruc Baci et al. 2024). Among these factors, the geometry input data, often extracted from DEMs, is known as one of the most significant inputs since it can change the propagation and the nature of flood in channel and adjoining floodplains (Hawker et al. 2018; Courty et al. 2019). DEMs can be derived from a wide range of sources which are different in resolution and cost, from high-resolution and accurate, but costly, such as LiDAR (Light Detection and Ranging), to coarse resolution and low-cost such as GDEMs derived by remote sensing techniques, including Advanced Spaceborne Thermal Emission and Reflection (ASTER), Shuttle Radar Topography Mission (SRTM), and Advanced Land Observing Satellite (ALOS). Although GDEMs offer valuable alternatives, particularly in regions lacking data, they come with several drawbacks. These include significant observational errors, a lack of fine spatial resolution, and the presence of noise, all of which contribute to uncertainty in the flood risk assessment (Aristizabal et al. 2024; Nandam and Patel 2024).

Numerous investigations have focused on evaluating the effectiveness of various GDEMs and comparing them with high-resolution DEMs such as LiDAR in modeling flood scenarios in river basins and non-urban areas (e.g., Md Ali et al. 2015; Saksena and Merwade 2015; Azizian and Brocca 2020). The flood inundation maps derived from LiDAR and UAVs (Unmanned Aerial Vehicles) with a higher horizontal and vertical resolution, are much more accurate compared to ones derived from GDEMs (Pasquier et al. 2019; Mohanty et al. 2020). Furthermore, results obtained in previous studies have also indicated that using ALOS and SRTM for simulation of floods in river basins performs better than DEMs derived by ASTER, and are comparable with those obtained with LiDAR (e.g.,Courty et al. 2019; Azizian and Brocca 2020; Khojeh et al. 2022).

In terms of urban flooding, Xu et al. (2021) investigated the role of DEM selection in urban flood simulations in Shanghai, utilizing six GDEMs. The results indicated that for all GDEMs, a reduction in spatial resolution led to an increase in both the predicted area of flood inundation and the maximum depth of inundation. In another study, Garrote (2022) analyzed flood modeling into Mocuba City, Mozambique, and concluded that Copernicus DEM model yielded more accurate results compared to SRTM, ALOS, and ASTER models. Conversely, Kramm and Hoffmeister (2022) noted that Copernicus DEM data could lead to ambiguous outcomes. Further, Fatdillah et al. (2022), focusing on Kuala Lumpur, Malaysia, found that the AW3D-5 DEM produced flood damage and risk estimates nearly twice as high as those from ALOS, and almost four times higher than those from SRTM. Additionally, Fereshtehpour and Karamouz (2018) employed

deterministic and probabilistic techniques to evaluate the effectiveness of multiple resampled LiDAR DEMs in flood inundation modeling. They concluded that resampled LiDAR DEMs are better suited than National Elevation Datasets (NEDs) for evaluating flood susceptibility in the Manhattan region of New York City, underscoring the crucial role of DEM selection in accurately simulating and predicting urban floods.

In addition to flood inundation maps, accurate topographic data is essential for reliable flood damage estimation and the creation of precise flood risk maps (Fereshtehpour et al. 2024; Nandam and Patel 2024). This importance stems from the direct correlation between the uncertainties in delineating flood extents and in quantifying key flood parameters (such as depth and velocity) with the integrity of flood risk assessments. Van de Sande et al. (2012) compared GDEMs with LiDAR-derived DEMs for assessing flood risk in a coastal area of Nigeria. Their findings indicated that GDEMs often lead to an underestimation of coastal flood risk due to the overestimation of land elevations. Nevertheless, previous studies have overlooked the influence of GDEMs in the assessment of flood hazards and flood damage within densely urbanized areas (e.g., McClean et al. 2020; Xu et al. 2021).

Besides evaluating GDEMs for flood modeling, resampling techniques are gaining increased focus in the field of urban flood modeling, aiming to balance the trade-off between accuracy and computational efficiency (Fereshtehpour and Karamouz 2018; Shen and Tan 2020). Saksena and Merwade (2015) resampled high-resolution LiDAR DEMs into a lower resolution one in a comprehensive study, and discovered a relation between the errors related to DEM attributes and the precision of fluvial flood inundation mapping. Brandt (2005) found that, by resampling the LiDAR DEM to lower resolutions, the spatial extent of floods increased as the resolution became coarser. This approach is particularly relevant in the context of choosing an appropriate DEM resolution for distributed hydraulic models.

The results of previous studies have underscored the importance of high-resolution DEMs for flood modeling. However, the availability of such DEMs remains one of the most crucial challenges in assessing flood inundation and, subsequently, flood damage in developing countries and data-sparse regions. While a wide range of studies have focused on the efficacy of GDEMs in estimating flood inundation in non-urban areas, but there is still a significant gap concerning their application in urban areas. This study addresses several novel contributions to the field:

1. Exploration of GDEM Effects in Urban Settings: While previous research has extensively investigated GDEMs in non-urban contexts, this study fills a critical gap by providing a comprehensive evaluation of GDEM performance in urban zones prone to flooding.
2. Assessment of ALOS and NASADEM Performance: This research is distinct in its evaluation of specific GDEMs, including ALOS-12.5, ALOS-30, and, most notably, NASADEM-30, alongside SRTM-30 and SRTM-90. The performance of NASADEM-30 in urban flood modeling has rarely been assessed.
3. Comprehensive Study on Flood Modeling: Prior studies lack a thorough analysis of flood inundation extents, damage estimations, and hazard mappings using GDEMs in urban contexts. Our study contributes a meticulous examination of these aspects, providing new insights and enhancing the understanding of GDEMs' utility in urban flood management phases.

Accordingly, this research aims to bridge the gap between the availability of high-resolution data and the practicality of lower-resolution models in managing urban floods, such as those in Tehran (Iran), which are highly susceptible to flooding (Tali and Nezammahalleh 2013). These floods often lead to extensive financial burdens, including damage to infrastructure, homes, and businesses, as well as significant personal losses. Assessing how DEMs, especially those of lower resolution, perform against a high-resolution reference map provides a pragmatic perspective on their utility. Such insights are not only beneficial for urban planners and disaster management authorities but also contribute significantly to the broader discourse on optimizing resource allocation for flood management in diverse urban contexts.

2 Materials and Methods

The methodological framework employed in this study is illustrated in Fig. 1, with further elaborations provided in the subsequent subsections.

2.1 Research Methodology

To achieve the stated aims, this study involved: 1) Collecting data, including hydraulic and hydrologic data, land use maps, economic data, and GDEMs with different spatial resolutions; 2) Evaluating the vertical accuracy of GDEMs against a high-resolution reference map; 3) Performing hydraulic modeling with MIKE FLOOD, a coupled

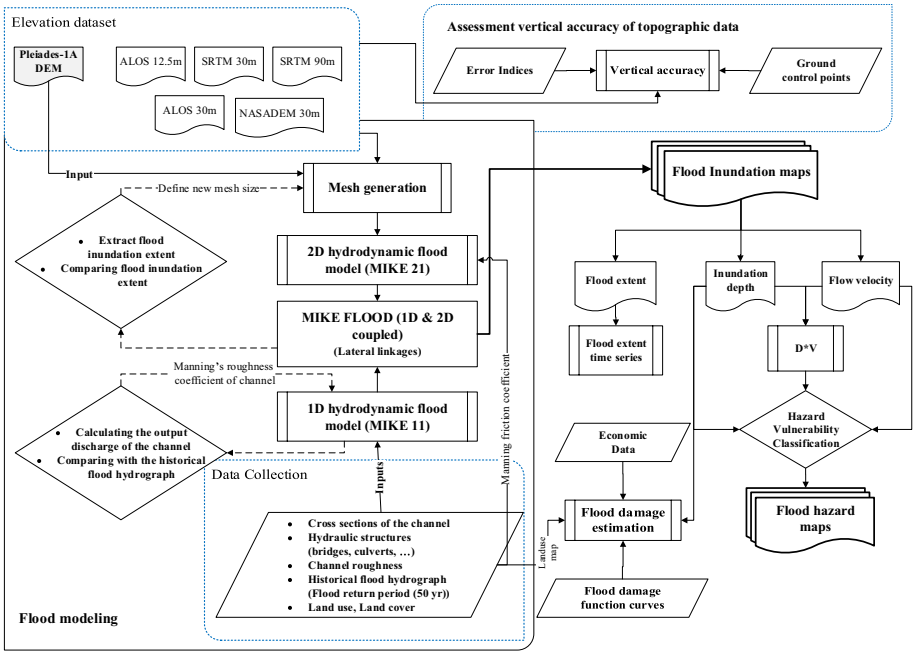


Fig. 1 Methodological flowchart for evaluating the impact of GDEMs on flood inundation mapping, hazard assessment, and damage estimation in an urban area

1D/2D flood model; 4) Creating flood inundation maps with various GDEMs; 5) Extracting flood hazard maps based on flood depth and flood velocity; and 6) Estimating flood damage using depth-damage curves.

2.2 Description of Study Area

The Flood Diversion Channel of West Tehran (FDCWT), approximately 9 km in length, functions as the primary drainage channel for surface runoff in Tehran's western region, covering a catchment area of 144.66 km^2, with 73.2 km^2 classified as urban areas. Located in an urbanized zone, its coordinates range from 35°43′1'' to 35°43′52'' N and from 51°17′27'' to 51°22′50'' E (Fig. 2).

The FDCWT, an east–west oriented channel with a rectangular profile, is constructed with concrete floors and walls made from a mixture of stone, sand, and cement mortar. Initiating south of Goftgoo Park, the FDCWT accumulates runoff from several subsidiary channels, including Tepehnizar, Farahzad, Shaghayegh, and Wesk, before eventually directing the flow into the Kan River. However, the channel's functionality has been compromised due to the unprincipled construction of 13 hydraulic structures along it, reducing its cross-sectional area. This issue, compounded by transformations in the surrounding land use, has diminished the channel's capacity, causing floodwaters to divert into urban spaces.

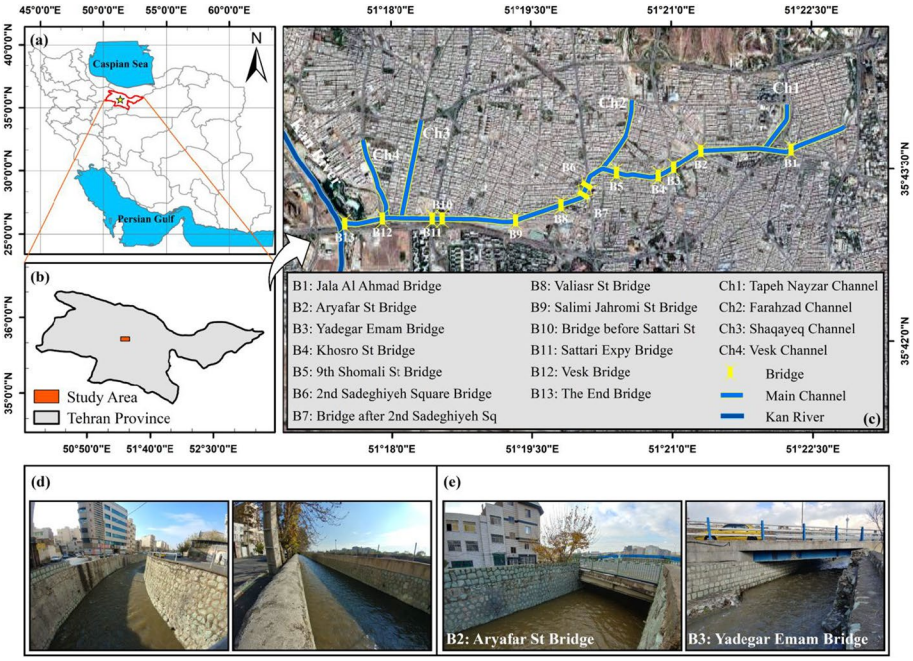


Fig. 2 Overview of the study area and the flood diversion channel in western Tehran, Iran. **a** and **b)** Show the location within Iran and Tehran Province. **c)** Illustrates the flood diversion channel and hydraulic structures. **d)** presents the channel's rectangular profile. **e)** Shows the two main bridges on the channel with sectional narrowing

2.3 Elevation Data Sources

In the present study, four distinct GDEM sources are examined: the commercial Pleiades-1A DEM, as the reference map (RM1), offering a 1-m resolution (De Lussy et al. 2012); ALOS PALSAR, available in 12.5 and 30-m resolutions (JAXA 2019); the SRTM data, provided in 30 and 90-m resolutions (Farr et al. 2007); and NASA's Digital Elevation Model (NASADEM) at a 30-m resolution (Bettiol et al. 2021). NASADEM-30 is the result of an exhaustive reanalysis of the initial SRTM DEMs, performed using enhanced algorithms and additional reference data. The specific attributes of these DEMs are detailed in Table 1.

2.4 Hydraulic Modelling

In this research, we employed MIKE FLOOD model to assess the influence of various GDEMs on flood characteristics such as depth, spatial extent, consequential damages, and hazard levels. MIKE FLOOD, a widely utilized flood modeling software, incorporates a 1D version (MIKE 11) for river channel modeling and a 2D version (MIKE 21) for floodplain inundation modeling.

MIKE 11, as a 1D hydrodynamic model, simulates one-dimensional, unsteady flow in rivers or channels, employing approaches such as kinematic, diffusive, dynamic, and full-dynamic wave simulations (Tansar et al. 2020). It calculates flow or water levels using an implicit, finite difference approach, based on a modified version of the Saint–Venant equations tailored for shallow water. The model assumes homogeneity, incompressibility, uniform velocity in the fluid, a water level parallel to the land surface, and minimal longitudinal hydrostatic pressure variation. The governing equations in MIKE 11 model are continuity and momentum equations, detailed in Eqs. (1) and (2), respectively.

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = q \tag{1}$$

$$\frac{\partial Q}{\partial t} + \frac{\partial\left(\alpha \frac{Q^2}{A}\right)}{\partial x} + gA\frac{\partial h}{\partial x} + \frac{gQ|Q|}{C^2AR} = 0 \tag{2}$$

where A is the cross-sectional area (m^2), Q is discharge (m^3/s), h is the stage above datum (m), q is the lateral inflow ($m^3/s/m$), C is the Chezy's roughness coefficient ($m^{1/2}/s$), R is the hydraulic radius (m), α is the momentum distribution coefficient ($s^2/$

Table 1 Summary of the global Digital Elevation Models (DEMs) evaluated in the present study

Dataset	Spatial resolution (m)	Method	Spatial coverage	Estimated vertical accuracy (m)	Released years
Pleiades-1A	1	photogrammetry	Global	0.3—0.5	2011
ALOS	12.5, 30	Stereo matching	82°N to 82°S	5	2016
SRTM	30	Interferometry synthetic aperture radar	60°N to 56°S	9	Since 2015
	90			16	2003
NASADEM	30	Interferometry synthetic aperture radar	60°N to 56°S	5	2020

m^3), g is the gravitational acceleration (m/s^2), x and t are the distance (m) and time (s), respectively.

MIKE 21 model is widely recognized 2D model for simulating fluvial, pluvial, and coastal flooding events, employs a numerical solution of full, time-dependent, and nonlinear conservation of mass and momentum equations based on depth-averaged Navier–Stokes equations (Mohanty et al. 2020; Tansar et al. 2020). It utilizes a finite difference method scheme to solve dynamic shallow water equations with constant grid spacing expressed as:

$$\frac{\partial z}{\partial t} + \frac{\partial p}{\partial x} + \frac{\partial q}{\partial y} = \frac{\partial d}{\partial t} \tag{3}$$

$$\frac{\partial p}{\partial y} + \frac{\partial}{\partial x}\left(\frac{p^2}{h}\right) + \frac{\partial}{\partial y}\left(\frac{pq}{h}\right) + gh\frac{\partial z}{\partial x} + \frac{gp\sqrt{p^2+q^2}}{C^2h^2} - \frac{1}{\varphi_w}\left[\frac{\partial}{\partial x}\left(h\tau_{xx}\right) + \frac{\partial}{\partial y}\left(h\tau_{xy}\right)\right] = 0 \tag{4}$$

$$\frac{\partial q}{\partial y} + \frac{\partial}{\partial y}\left(\frac{q^2}{h}\right) + \frac{\partial}{\partial x}\left(\frac{pq}{h}\right) + gh\frac{\partial z}{\partial y} + \frac{gp\sqrt{p^2+q^2}}{C^2h^2} - \frac{1}{\varphi_w}\left[\frac{\partial}{\partial y}\left(h\tau_{yy}\right) + \frac{\partial}{\partial x}\left(h\tau_{xy}\right)\right] = 0 \tag{5}$$

where z is surface elevation (m), p is flux density in the x-direction ($m^3/s/m$), q is the lateral inflow (m^3/s), h is water depth (m), t is time (s), x and y are space coordinates (m), C is the Chezy resistance, w is the density of water (kg/m^3), τ_{xx}, τ_{xy} and τ_{yy} are the components of effective shear stress (N/m^2), and g is acceleration due to gravity (m/s^2).

In this study, the initial stage of flood modeling involved simulating the main channel's flow with MIKE 11, based on channel topography data sourced from the Tehran Municipality. This data includes detailed information on the channel's cross-sectional configuration, slope, Manning's roughness coefficient, and the specifications of hydraulic structures along the channel. Subsequently, a 50-year return period flood hydrograph, derived from a hydrological model (HEC-HMS) and documented in previous study of this specific area (Yavari et al. 2022), was integrated as the boundary condition for the simulation. The boundary conditions for the FDCWT simulation includes upstream, downstream, and intermediate channel parameters. MIKE 11 effectively predicts flood parameters such as water depth and discharge along the channel but lacks the capability to model flood extent into floodplains. Therefore, MIKE 21 was utilized to simulate floodplain inundation, and its integration with MIKE 11 enables 2D flow simulation, enhancing flood behavior accuracy in floodplains. The key input for MIKE 21, the bathymetry layer, was derived from the GDEMs, with RM1 as the reference map. To enhance the simulation's accuracy and performance, a mesh independence study was conducted, resulting in the selection of optimal mesh size (5 m^2) for bathymetry extraction. Additionally, distributed Manning's resistance values were applied to reflect different land use impacts on flood dynamics.

Ultimately, we coupled both MIKE 11 and MIKE 21 models utilizing MIKE FLOOD based on the lateral link, enabling dynamic links between the river network and floodplain's bathymetry. Lateral links effectively join the FDCWT's river network to the floodplain bathymetry, using a cell-to-cell method to allow flow overflow into the floodplain. It is worth noting that the MIKE 11 model remained consistent in the coupling process with MIKE 21 models constructed using various GDEMs.

2.5 Flood Risk Assessment

The risk assessment and subsequent analysis were concentrated on the region delineated in Fig. 4a, for two principal reasons. Initially, this area experienced the most extensive flooding, as depicted in the reference inundation map (Fig. 4a), attributed to the inadequate performance of the hydraulic structures, specifically the Aryafar St. Bridge and Yadegar Emam Bridge (Fig. 2e). Additionally, this region exemplifies a high-density urban area, aligning with the objectives of the study.

Flood inundation maps were evaluated based on several criteria: flood extent and depth, associated damages, and the level of flood hazard. For comparison, the inundation map generated from the RM1 served as the benchmark. The RM1 has high horizontal resolution and vertical accuracy (Zhou et al. 2015; Bagnardi et al. 2016), and is validated using ground control points provided by Tehran Municipality. The following sections detail the methods used for hazard mapping and estimating flood damages.

2.5.1 Flood Hazard Mapping

Flood hazard is defined as the threat represented by a flood of certain frequency and magnitude to a particular location. The assessment of flood-hazard variability within a flood-vulnerable urban zone is essential for management and description of danger to property, infrastructure, and people. Flood hazard is commonly assessed based on the outcome of physically based models that simulate the water movement across the floodplain (Teng et al. 2017). The primary parameters used to measure flood hazard include flood depth (Beadenkopf 2013) and flow velocity (Kreibich et al. 2009), either considered individually or combined to establish an appropriate flood hazard indicator. In this research, we utilized the flood-hazard index classification approach from AIDR (2017) and Smith et al. (2014), which merges flood depth and flow velocity parameters, to illustrate the extent of flood risk. The categorized inundation depth (D) and flood velocity (V) are outlined in Table 2 to simplify the hazard classification. For the utmost flood hazard potential, a specific combination of flood depth and flow velocity (DV) was identified to depict the community's flood risk accurately. The flood-hazard classification resulted in six distinct classes: H1 through H6.

2.5.2 Flood Damage Estimation

Evaluating the efficacy of GDEMs in estimating flood damage is imperative given that the spatial variability in flood hazards, attributed to DEMs of diverse resolutions, significantly influences the assessment of flood damage within urban areas. Given the current absence of specific depth-damage data for urban assets in Tehran, we employed the depth-damage curves of the US Army Corps of Engineers (USACE U.S. Army Corps of Engineers 2003) for various land uses to estimate the flood damages, as recommended in guidelines from Iran's "Flood Damage Consideration" (Ministry of Energy 2016). These depth-damage functions encompass a broad range of building characteristics which align closely with the diversity of structures within Tehran, including factors like the number of floors, structural framework, and age of the buildings. This alignment ensures a more accurate estimation of potential damage in the urban context of Tehran

where specific local depth-damage data is not available. This study focuses on three main types of land uses, including commercial, residential, and service (infrastructure) buildings, each with its own depth-damage function and economic value (replacement cost). Delving into specifics, to estimate flood damages, floodplain and land use layers are overlaid in the Geographic Information System (GIS) Environment and used as inputs for the depth-damage functions. Indeed, after flooding occurs, flood depth is entered into the damage function, from which the damage percentage is obtained. This output is then multiplied by the economic value of that building to estimate the induced damage. It is important to note that road infrastructure is not included in flood damage estimation. Our study primarily focuses on the impacts of GDEMs on urban flooding, specifically emphasizing flood extent and depth, not velocity. This focus is supported by the fact that velocity predominantly affects road infrastructures rather than residential buildings (Kreibich et al. 2009; FEMA 2012; Fereshtehpour and Karamouz 2018), therefore, our analysis concentrates solely on evaluating flood damage to buildings.

2.6 Resampling

Utilizing maps with lower resolution in flood studies indeed offers certain advantages, particularly in terms of computational and financial aspects. Considering the high costs associated with acquiring high-resolution DEMs, such as those produced by LiDAR technology, identifying an optimal resolution that balances accuracy with affordability is crucial—particularly for resource-constrained settings often encountered in developing countries. In this study, to assess the trade-off between precision and computational efficiency of DEM resolution for urban flood simulation, the high-resolution reference map (RM1) is resampled into a coarser DEM with a spatial resolution of 5 m (Res_5 m) using three resampling techniques namely, nearest neighbor, bilinear and cubic convolution. The results of these three methods were then compared to the RM1 in terms of error statistics. The analysis revealed that bilinear interpolation exhibits superior performance compared to alternative approaches (e.g., Fewtrell et al. 2008; Fereshtehpour and Karamouz 2018; Muthusamy et al. 2021). It is important to note that detailed data regarding this comparison is not included, as they fall outside the scope of the study's main objectives. Subsequently, the efficacy of Res_5 m is assessed in flood inundation mapping, as well as in evaluating flood damage and hazard mapping.

2.7 Evaluation Indices

To assess the vertical accuracy of different GDEMs over the study area (Fig. 4a), the RM1 was employed as the benchmark. A total of 2,000 control points were extracted from the RM1 to facilitate the evaluation of vertical accuracy. For this analysis, quantitative indices including Root Mean Square Error (RMSE), Mean Error (ME), Standard Deviation (SD), and Correlation Coefficient (R^2) were employed. RMSE is a widely utilized statistic that accommodates both variance and bias error. ME measures the systematic error indicating the degree to which the data set is, on average, underestimated or overestimated compared to the ground control points. SD quantifies the

Table 2 Combined hazard curves – vulnerability thresholds classification limits

Hazard Vulnerability Classification	Description	Classification limit (D and V in combination) m^2/s	Limiting still water depth (D) m	Limiting velocity (V) m/s
H1	Generally safe for vehicles, people, and buildings	D*V < = 0.3	0.3	2
H2	Unsafe for small vehicles	D*V < = 0.6	0.5	2
H3	Unsafe for vehicles, children, and the elderly	D*V < = 0.6	1.2	2
H4	Unsafe for vehicles and people	D*V < = 1.0	2	2
H5	Unsafe for vehicles and people. All building types are vulnerable to structural damage. Some less robust building types vulnerable to failure	D*V < = 4.0	4	4
H6	Unsafe for vehicles and people. All building types considered vulnerable to failure	D*V > 4.0	-	-

random error of the data set and shows the dispersion of the errors around its average. R^2, on the other hand, is a statistical measure used to determine how well a model fits a data set. Equations (6) to (9) delineate the characteristics of these statistical errors.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}\left(Z_{DEM\,i} - Z_{DEM_r}\right)^2}{N}} \tag{6}$$

$$ME = \frac{\sum_{i=1}^{N}\left(Z_{DEM\,i} - Z_{DEM_r}\right)}{N} \tag{7}$$

$$SD = \sqrt{\frac{\sum_{i=1}^{N}\left(\left(Z_{DEM\,i} - Z_{DEM_r}\right) - ME\right)^2}{N-1}} \tag{8}$$

$$R^2 = \frac{\frac{1}{N}\sum_{i=1}^{N}\left(Z_{DEM\,i} - \overline{Z}_{DEM\,i}\right)\left(Z_{DEM\,r} - \overline{Z}_{DEM\,r}\right)}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(Z_{DEM\,i} - \overline{Z}_{DEM\,i}\right)^2} * \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(Z_{DEM\,r} - \overline{Z}_{DEM\,r}\right)^2}} \tag{9}$$

$$F = \frac{A_{DEM_r} \cap A_{DEM_i}}{A_{DEM_r} \cup A_{DEM_i}} \tag{10}$$

where Z_{DEM_r} denotes the elevation value derived from the RM1. Conversely, Z_{DEMi} represents the corresponding elevation value sourced from other GDEMs. The terms $\overline{Z}_{DEMr}$ and $\overline{Z}_{DEMi}$ are the averages of the elevation values from the RM1 and GDEMs, respectively. N stands for the total count of points from which elevation values are extracted.

These functions are also used to quantify errors in estimated flood depth. However, to evaluate the accuracy of the simulated flood extent and to determine the similarity of the predicted flood extent from GDEMs to the reference map, the goodness-of-fit measure (F) is used (Bates and De Roo 2000), as presented in equation (10). Where A_{DEM_r} is the flood inundation area simulated with the RM1, and A_{DEM_i} the flood inundation area simulated with other GDEMs.

3 Results

3.1 Assessment Vertical Accuracy of Topographic Data

As shown in Fig. 3, NASADEM-30 demonstrates the highest vertical accuracy in the study area, recording the lowest RMSE of 6.48 m and a minimal ME of 0.18 m. Following closely, SRTM-30 reports an RMSE of 6.70 m, but a ME of -0.89 m indicates a slight underestimation in the vertical measurements. In contrast, the medium-resolution ALOS-12.5 registers an RMSE of 6.82 m and a ME of 1.44 m, indicative of a slight overestimation, offering a counterpoint to SRTM-30's underestimation. Other datasets, including ALOS-30 and SRTM-90, show more significant ME discrepancies, suggesting potential

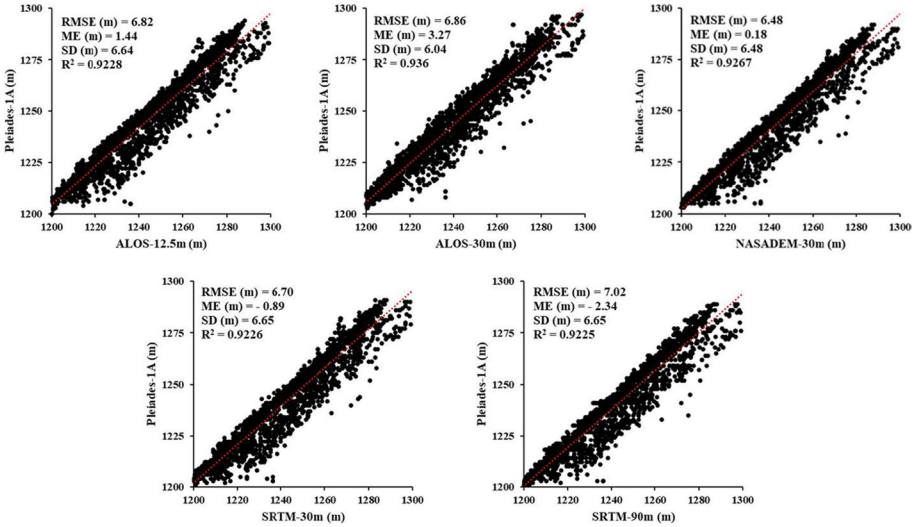


Fig. 3 Evaluation results of the vertical accuracy of Global Digital Elevation Models (GDEMs) against the reference map

biases in vertical accuracy. The standard deviations are consistently within 6–7 m for all datasets. Conclusively, the determination coefficient (R2) values range from 0.9225 to 0.936, indicating a strong correlation with the RM1.

3.2 Effect of Using Different GDEMs on Flood Inundation

Figure 4 depicts the impact of using GDEMs from different sources on hydraulic modeling, highlighting variations in flood inundation extent and depth across the urban area. This area is divided into three zones—Zone-1, Zone-2, and Zone-3—with radii of 900 m, 1400 m, and 2000 m, respectively, originating from the channel's leakage point towards the urban center.

3.2.1 Flood Inundation Extent

The flood inundation extents derived from ALOS-12.5 and ALOS-30 demonstrate constraints in accurately simulating the inundation domain in Zones 2 and 3 (Fig. 4c and d). In contrast, models based on NASADEM-30, SRTM-30, and SRTM-90 closely match the RM1 across all zones, although SRTM-90 slightly extends beyond Zone 3's boundaries.

Decreasingthe spatial resolution in GDEMs may omit critical topographical details, leading to overestimation or underestimation of flood extents compared to the RM1, which captures finer topographical nuances due to its higher resolution. Figure 5 also illustrates the time series analysis of inundation areas derived from various GDEMs, highlighting discrepancies in flood extent over time. Notably, SRTM-90, SRTM-30, and NASADEM-30 overestimate the flood areas, whereas ALOS-30 underestimates them. Although ALOS-12.5 aligns with the RM1 in the time series, spatial inaccuracies in flood extent are significant, as shown in Fig. 4c.

3.2.2 Flood Depth Distribution

To illustrate the flood depth distribution within the inundated regions, the flood depth was categorized into five ranges as shown in Fig. 4a. The flood inundation map from the RM1 demonstrates varied flood depths across the area. Notably, nearly 47% of the area

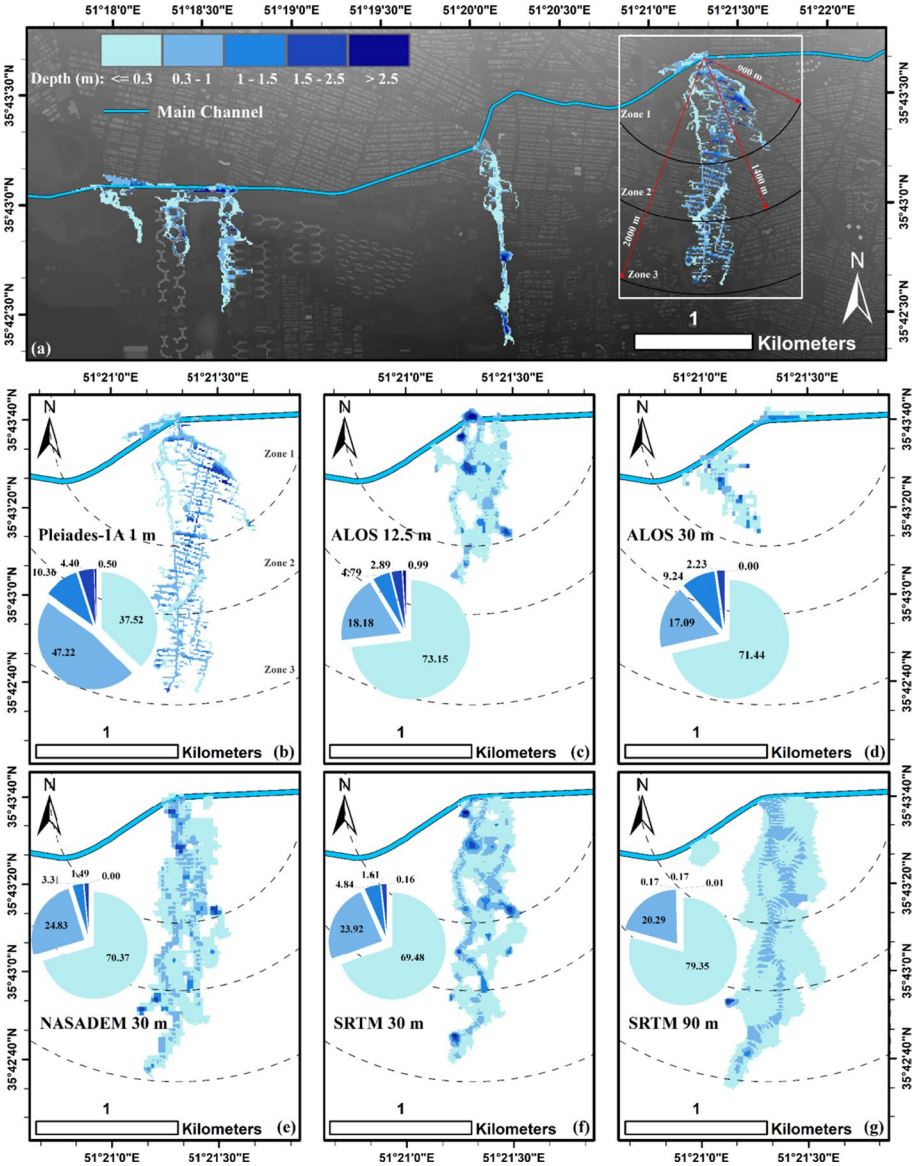


Fig. 4 Comparative visualization of flood inundation extent derived from various GDEMs and a high-resolution reference map. **a)** Depicts the overall flood extent along the channel. Subsequent panels illustrate the inundation extent within a focused section of panel a, with **b)** Using the reference map (Pleiades-1A 1m), **c)** ALOS-12.5 m, **d)** ALOS-30 m, **e)** NASADEM-30 m, **f)** SRTM-30 m, and **g)** SRTM-90 m

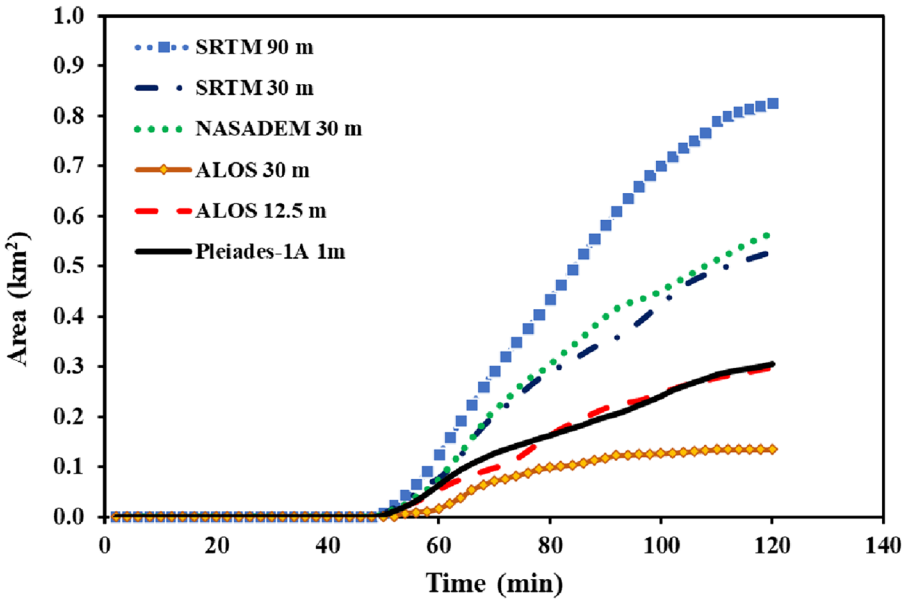


Fig. 5 Time series analysis of flood inundation areas generated from various GDEMs, compared to a high-resolution reference map

experiences a flood depth of 0.3 to 1 m while around 37.5% has depths of less than 0.3 m. Additionally, depths of 1 to 1.5 m and 1.5 to 2 m cover approximately 10% and 4.5% of the region, respectively. Unlike the RM1, flood depths derived from GDEMs show a more uniform distribution, with a significant portion, about 70%, exhibiting depths of less than 0.3 m—nearly twice the percentage observed in the RM1.

As mentioned before, NASADEM-30 and SRTM-30 excel in reproducing the flood extent observed in the RM1. Focusing on NASADEM-30, it is evident that only about 25% of the flood depth ranges from 0.3 to 1 m, which is roughly half of what is registered in the RM1. Conversely, regions with flood depths below 0.3 m are almost double the proportion found in the RM1, as illustrated in Fig. 4e. This discrepancy underscores a critical insight: even when GDEMs closely match the RM1 in flood extent, they may significantly differ in depth distribution, often overestimating shallower zones and underestimating deeper inundations.

3.2.3 Flood Extent and Depth Uncertainty

In evaluating the uncertainty of predicted flood extents, we analyzed both the variance in flood extent areas and the congruence percentage of flood extents using F-statistics, enabling a comparison among various GDEMs (Table 3). Given the RM1 as the reference map, comparisons revealed significant differences. ALOS-30 shows an inundation area considerably smaller by 55% compared to the RM1. In contrast, NASADEM-30, SRTM-30, and SRTM-90 projected increasingly larger areas, exceeding the RM1 by 83%, 71%, and 167%, respectively. These results indicate that decreasing DEM resolution, the estimated flood extent increases.

Table 3 Comparative depth statistics and performance metrics for flood inundation analysis using different GDEMs versus a high-resolution reference map (Pleiades-1A 1m)

Dataset	Depth statistics (m)		Performance metric				
	max	mean	RMSE (m)	ME (m)	SD (m)	F (%)	Area (km^2)
Pleiades-1A 1m	3.58	0.55	-	-	-	-	0.31
NASADEM-30 m	3.38	0.31	0.63	-0.29	0.56	28.72	0.57
ALOS-12.5 m	3.4	0.34	0.89	-0.27	0.85	17.68	0.3
ALOS-30 m	2.56	0.38	0.50	0.33	0.38	2.14	0.14
SRTM-30 m	2.4	0.30	0.67	-0.27	0.61	28.03	0.53
SRTM-90 m	2.22	0.21	0.62	-0.36	0.50	25.50	0.83

Although ALOS-12.5 closely aligns with the RM1 in overall inundation extent area, showing an only -3% difference, it inaccurately represents southern flood zones (Zones 2 and 3). This limitation is reflected in its F-Statistics at 17.68% lower than NASADEM-30 (28.72%), SRTM-30 (28.03%), and SRTM-90 (25.50%). The primary causes for this mismatch in inundation extents, as suggested by F-Statistics, are the reduction in spatial resolution, and differences in data acquisition methodologies, resulting in diminished bathymetric accuracy.

Regarding the flood depth uncertainty, NASADEM-30 and SRTM-30, closely aligning with the RM1 in flood extent, exhibit similar metrics. These GDEMs' flood depth predictions diverge from the RM1 by an average of ±0.65 m, consistently underestimating by -0.27 m to -0.29 m. Variability in predictions (SD values) is 0.56 m for NASADEM-30 and 0.61 m for SRTM-30. Despite their similarities, their performance diverges when focusing on depth: RM1's average depth is 0.55m, while these GDEMs indicate a 45% reduction. Furthermore, SRTM-30 significantly underrepresents the highest flood depth by 33% compared to the RM1, whereas NASADEM-30 indicates a peak depth of 3.38 m, exceeding SRTM-30 nearly 1 m.

SRTM-90, compared to NASADEM-30 and SRTM-30, shows similar RMSE, ME, and SD results but is less effective in accurately representing flood extent due to its coarser resolution. Its mean flood depth is about 62% lower than that of the RM1, and it records a maximum flood depth of approximately 2.22 m, which is 1.4 m less than the RM1's. These findings highlight the limitations of SRTM-90 in providing detailed and accurate flood depth measurements.

Results reveal that lower-resolution GDEMs, such as NASADEM-30 and SRTM-30, often lead to an overestimation of the flood extents and an underestimation of flood depths. Our analysis, leveraging specific error metrics and F-Statistics, suggests that ALOS-12.5 and ALOS-30 underperform in accurately modeling urban flood inundation. SRTM-90 notably overestimates flood extent, leading to a simulated floodplain that poorly matches observed conditions. Among the evaluated GDEMs, NASADEM-30 stands as the most effective in replicating both the extent and depth of flood inundation in our study area.

3.3 Effect of GDEMs on Flood Hazard Mapping

To assess the impact of different GDEMs on flood hazards, we employed a flood hazard index classification methodology that considers both depth and flow velocity. Figure 6 presents the hazard map for FDCWT, derived from different GDEMs. The results show the

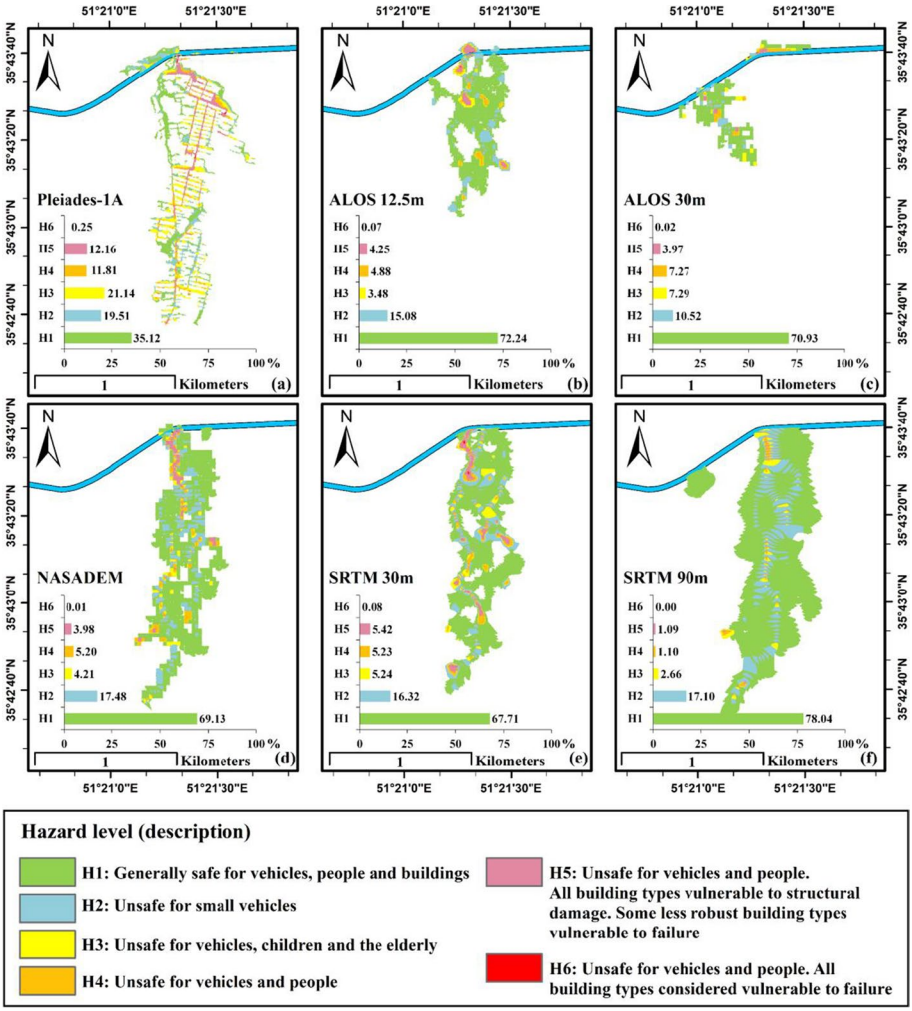


Fig. 6 Flood hazard maps derived from various GDEMs compared to a high-resolution reference map. **a)** Reference map (Pleiades-1A), **b)** ALOS-12.5 m, **c** ALOS-30 m, **d)** NASADEM-30 m, **e)** SRTM-30 m, **f)** SRTM-90 m, depicting hazard levels from H1 (generally safe) to H6 (considered vulnerable to failure)

accuracy in extracting flood parameters like depth and velocity directly enhances the flood hazard distribution map, ensuring it more precisely reflects flood consequences. This accuracy is especially evident in the map from the RM1, which outperforms other GDEMs in depicting flood depth and extent, as shown in Fig. 4b.

According to Fig. 6a, the RM1 shows that 35.12% of the inundation area falls under category H1, indicating zones largely safe for vehicles, people, and structures. Categories H2 and H3, denoting increased risk zones, collectively cover about 40.5% of the area; 19.5% in H2 pose risks for smaller vehicles, and H3 areas extend hazard to larger vehicles, children, and the elderly. The remaining categories constitute roughly 25% of the area with H4 (11.81%) highlighting dangers for vehicles and individuals. H5 (12.6%) posing risks

to infrastructure's stability and safety, and H6 (0.25%), the most critical zone, indicating imminent catastrophic failure for nearly all building types.

Figure 6d presents the flood hazard map from NASADEM-30, highlighting uncertainties in flood depth and inundation extent due to increased spatial resolution, which impacts flood hazard quantification. Notably, about 69% of the area is categorized as H1, twice the percentage of the RM1, indicating an expansion with shallow floods (depths under 0.3 m) as vertical accuracy decreases. Conversely, categories H4, H5, and H6 together account for only 10%, half that of the RM1, suggesting a decrease in areas with deeper floods using NASADEM-30. Categories H2 and H3 represent about 17% and 4%, respectively, showing a decrease from RM1's 19.5% for H2 and 21% for H3. SRTM-30, slightly less accurate in flood mapping than NASADEM-30, shows a similar flood hazard pattern (Fig. 6e).

As outlined in Section 3.2, GDEMs sourced from ALOS-12.5, ALOS-30, and SRTM-90 exhibit limitations in accurately simulating flood inundation extent and depth. Consequently, the flood-hazard maps derived from these GDEMs not accurately reflect the actual flood hazard scenario, as demonstrated in Fig. 6b, c, and f.

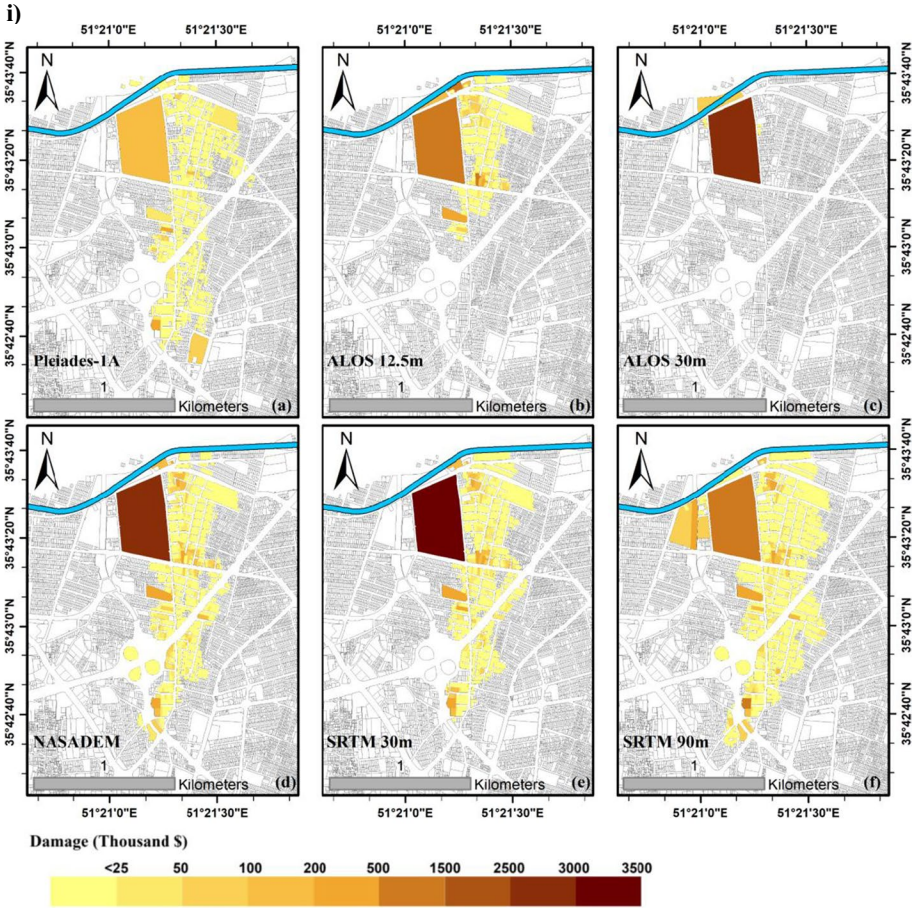
Comparing across different GDEMs reveals that a decrease in spatial resolution correlates with an increase in category H1 areas on the flood hazard map; a lower resolution leads to a larger area being classified as low risk (H1). Among the evaluated GDEMs, NASADEM-30 and SRTM-30 emerge as superior for estimating flood hazard maps, outperforming ALOS-12.5, ALOS-30, and SRTM-90.

3.4 Effect of GDEMs on Flood Damage Estimation

To estimate flood damages to primary land uses, including commercial, residential, and service buildings, we employed the depth-damage curves recommended by the US Army Corps of Engineers (USACE U.S. Army Corps of Engineers 2003). Figure 7i depicts the spatial distribution of flood damages across the study area for a 50-year flood event. Figure 7ii further explores how GDEMs affect flood damage assessments, comparing damage across various land uses with deviations from the RM1. The outcomes indicate that differences in GDEMs lead to variation in the extent and pattern of flood damage, largely due to the precision of flood depth calculations and distribution identified in flood simulations.

Based on the RM1, the total estimated flood damage is $8,662,000, with residential areas bearing the burnt at $6,913,000 (79.8% of the total), highlighting their significant flood vulnerability. Commercial and Infrastructure sectors faced damages of $1,005,000 (11.6%) and $745,000 (8.6%), respectively. Compared to the reference, NASADEM-30 and SRTM-30 consistently estimate higher damages across land use due to decreased spatial resolution, resulting in overestimations. For instance, NASEDEM-30 estimates total damage at approximately $22.5 million, about 160% above the RM1, with residential, commercial, and infrastructure damages significantly higher by $8.22 million, $2.41 million, and $3.36 million, respectively. Figure 7i underscores the substantial discrepancies in damage assessment when using NASADEM-30, pointing to the necessity for high-resolution data to improve the accuracy of flood impact evaluations. Like NASADEM-30, the SRTM-30 dataset similarly overestimates the flood damage across all categories. The key distinction, however, is that SRTM-30 tends to overestimate to a greater extent than NASADEM-30, indicating that while both datasets may closely match the reference map's flood extent, their damage estimations vary, with SRTM-30 projecting even higher potential losses. The total overestimation by SRTM-30 is slightly above 10% compared to the already elevated estimates provided by NASADEM-30.

i)



ii)

Land use	Flood damage (Thousand$)					
	Pleiades-1A	ALOS 12.5 m	ALOS 30 m	NASADEM 30 m	SRTM 30 m	SRTM 90 m
Residential	6913	6926	67	15141	15380	18450
Commercial	1005	2668	75	3410	3414	4212
Infrastructure and public services	745	2416	2840	4106	4754	2955
Total	8662	12010	2982	22658	23548	25617

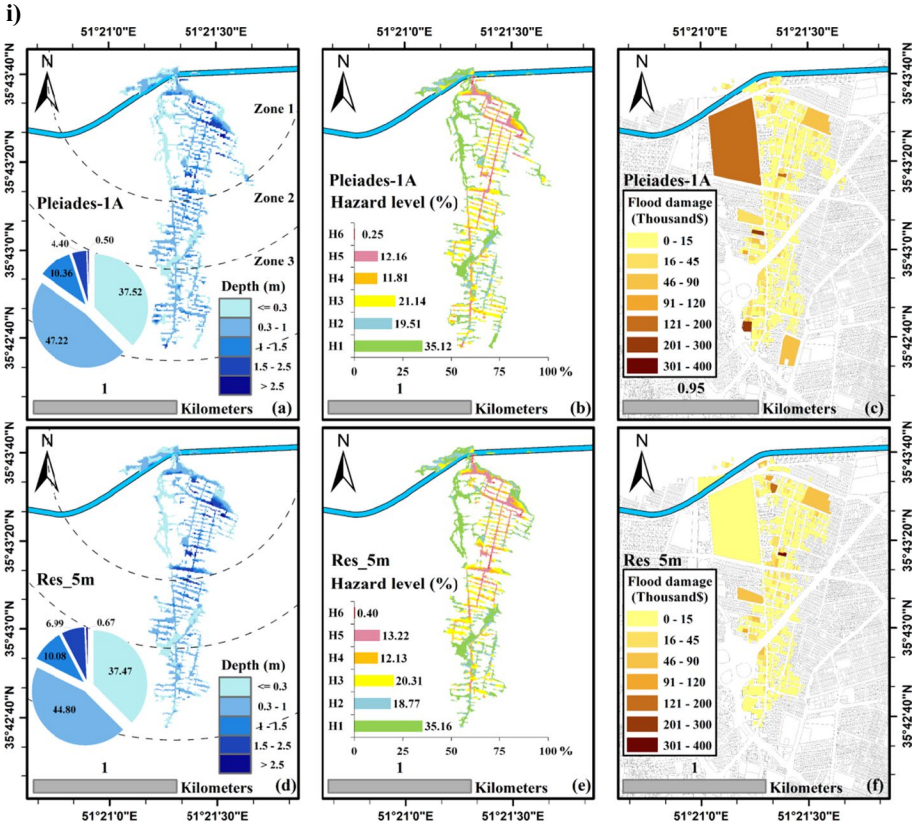
Fig. 7 (i) Flood damage estimation maps utilizing GDEMs and a high-resolution reference map. **a)** Reference map (Pleiades-1A), **b)** ALOS-12.5 m, **c)** ALOS-30 m, **d)** NASADEM-30 m, **e)** SRTM-30 m, **f)** SRTM-90 m, with damage quantified in thousands of dollars across the affected areas. (ii) Comparative analysis of flood damage by land use category across GDEMs and the reference map

Furthermore, SRTM-90 records the highest damage estimates among all evaluated datasets at $25.6 million, nearly four times the estimate of the RM1. This overestimation stems from its coarser resolution, which leads to a more generalized flood depth and wider extent, predominantly under 0.3 m depth, thereby overstating the impact area. Conversely, ALOS-12.5 and ALOS-30 struggle to precisely delineate flood boundaries, significantly underestimating flood extents and failing to accurately map the spatial distribution of damages. As a result, their damage estimates considerably lack accuracy compared to the RM1, with ALOS-30 underestimating total damages by 65%.

3.5 Resampling

To assess the trade-off between precision and computational efficiency in DEM resolution for urban flood simulation, we employed the bilinear interpolation technique to create a coarser 5-m

i)



ii)

Dataset	Performance metric					
	RMSE (m)	ME (m)	MAE (m)	SD	F (%)	Area (km^2)
Res_5 m	0.29	0.06	0.17	0.28	73.11	0.29

Dataset		Land use			
		Residential	Commercial	Infrastructure and public services	Total
Pleiades-1A 1m	Flood damage (ThousandS)	6913	1005	745	8662
Res_5 m		4936	896	492	6324

Fig. 8 (i) Comparison of flood impact assessments between the high-resolution reference map (Pleiades-1A) and the resampled 5-m map (Res_5 m): (**a**) and (**d**) Show flood inundation extent, (**b**) and (**e**) Indicate flood hazard levels, and (**c**) and (**f**) Display estimated flood damages in thousands of dollars. (ii) Evaluation of the Res_5 m DEM performance metrics and flood damage across different land use categories, compared to the reference map

resolution DEM (Res_5 m) from the RM1. Figure 8i presents the Res-5 m results for flood inundation extent, hazards, and damages, comparing them to the RM1 outcomes.

The Res_5 m model covers all area simulated by the RM1 but faces challenges in capturing fine details, particularly in Zones 1 and 3, as shown in Fig. 8i-a and i-d. The F-statistic of approximately 73% confirms a high similarity in flood extent with the RM1. The spatial distribution of flood depth in Res_5 m also closely matches the RM1's pattern: about 45% of the region has flood depths of 0.3 to 1 m, slightly less by 2% than the RM1, while depths under 0.3 m and between 1 to 1.5 m are similar, accounting for 37% and 10% respectively. Depths of 1.5 to 2 m cover 4.5%, 3% less than reference. Error indices for Res_5 m (Fig. 8ii) reveal an RMSE of 0.29 m from the RM1, indicating prediction error magnitude, and an ME of 0.06 m suggests minor underestimation in depths. The MAE and SD, at 0.17 m and 0.28 m respectively, reflect average differences and variability in predictions. Overall, Res_5m's predicted inundation area is around 0.29 km^2, 6% less than the RM1's 0.31 km^2, likely due to the smoothing effect of elevation resampling. Such smoothing typically leads to an overall increase in land elevation, aligning with the outcomes reported in prior research (Fereshtehpour and Karamouz 2018).

Figure 8i-b and i-e show the flood hazard maps derived from the RM1 and Res_5 m DEMs, respectively. The analysis indicates that reducing spatial resolution to 5 m does not significantly alter the flood hazard map, attributed to the satisfactory overlap of flood zones and the consistent distribution of parameters like depth and velocity. Most of the inundation area falls under H1 and H3, comprising about 35% and 20% of the area, respectively, with a negligible difference of about 1% from the RM1, as shown in Fig. 8i-e. Likewise, the combined severe categories, H4, H5, and H6, account for about 26%, consistent with the RM1. These results affirm that the Res_5 m DEM effectively generates the hazard map for the study area.

Figure 8i-c and i-f illustrate the spatial resolution of flood damages from coarsening the pixel size from 1 to 5 m. Furthermore, Fig. 8ii details the damages across various land use and compares them to the RM1 outcomes. The findings indicate that Res_5m's smaller flood inundation extent, relative to the RM1, leads to underestimation flood damage, with total damages for Res_5 m reducing by approximately 27% to $6,324,000. Especially, residential zone damages in Res_5 m are about 28% lower than the RM1's estimates. For commercial and infrastructure land uses, the reductions are approximately 10% and 34%, respectively.

Given the high costs and processing challenges of using high-resolution commercial DEMs globally for hydraulic modeling, assessing lower-resolution DEMs' performance in flood simulation becomes crucial, including their effectiveness in hazard and damage mapping. The insights from this section suggest that where high-resolution maps are unavailable, lower-resolution alternatives (e.g., with 5-m spatial resolution) can reliably delineate flood zones, generate hazard maps, and estimate flood damages, particularly in densely urbanized areas.

4 Discussion

Given the results of this study, it is evident that the accuracy of flood inundation mapping, flood hazard mapping, and flood damage estimation decreases in densely urban areas when using GDEMs instead of high-resolution topographic data. Indeed, the overestimation of flood extent and the underestimation of flood depth are identified as the main impacts of utilizing GDEMs, leading to inaccurate flood risk assessments in terms of flood hazards and damages. Therefore, although GDEMs provide valuable data, especially for data-scarce

areas, their use for risk assessment in urban settings necessitates additional caution due to their potential for negative consequences.

In terms of flood extent, however, most of GDEMs had a spatial resolution of 30 m, their performance in flood inundation mapping was different, attributable to their data acquisition and processing methodologies, spatial resolution, and inherent errors (Wise 2000; Fisher and Tate 2006; Wechsler 2007). Given the results, NASADEM-30 and SRTM-30 produce approximately the same spatial distribution pattern of the flood extent. However, they are incapable of capturing detailed flooding at street scales, leading to an overestimation of flooded areas (e.g., Xu et al. 2021). In contrast, ALOS DEMs significantly underestimate the flood extent, resulting in an inaccurate flood extent pattern compared to the RM1. This result also highlights the point that the lower-resolution GDEMs, here ALOS-12.5, necessarily cannot provide an acceptable flood extent.

Regarding the acquisition methodologies, NASADEM-30 and SRTM-30, derived using the Interferometric Synthetic Aperture Radar (InSAR) method, can penetrate cloud cover and vegetation, offering a more precise representation of ground surface. This contrasts with ALOS, which relies on Stereo Matching using optical imagery, less effective under vegetation or cloudy conditions and potentially overestimating elevation (Muench et al. 2022). The optical imagery's limitation is significant, representing the tops of trees as ground elevation, whereas radar-based methods provide a more accurate ground level by partially penetrating the tree canopy. The vertical accuracy assessment also confirmed this, from which the RMSE of the ALOS DEMs was approximately 7 m, which is higher than the generally estimated RMSE of around 5 m (Tadono et al. 2014). Also, in contrast to other GDEMs, ALOS-12.5 and 30 overestimated the elevation values by around 1.5 and 3.5 m, respectively, based on the ME index.

The low spatial resolution of these models, typically 30 m, also exacerbates the issue by smoothing over critical small-scale features like buildings, roads, and drainage channels, leading to an incapacity to accurately depict street-scale features in urban environments, resulting in significant positive elevation discrepancies (Rodriguez et al. 2006; Sampson et al. 2016). These constraints also contribute to underestimation of flood depth when using GDEMs. The smoothing effect regarding the low spatial resolution, especially in urban areas, and the positive bias of GDEMs lead to flood depth underestimation. Consequently, in this case study, in SRTM-30 and NASADEM-30 datasets, approximately 70% of the flood extent—nearly double the percentage observed in the RM1—was categorized with a depth of less than 30 cm. Furthermore, approximately 24% of the flood extent had a flood depth between 0.3 m and 1 m, which is roughly half compared to the RM1. The remaining portion, with a depth greater than 1 m, was only 5%, indicating an underestimation of the deeper flood extent.

This underestimation of flood depth was also evident in ALOS, despite their underestimation of the overall flood extent. This discrepancy underscores the notion that while some GDEMs might align closely with the RM1 in terms of flood extent, there can be considerable variations in depth distributions, often over-representing shallower zones and under-representing deeper inundations. The previous studies, both in rural (e.g., Fatdillah et al. 2022; Garrote 2022) and urban areas (e.g., Muthusamy et al. 2021; Xu et al. 2021), confirm our results in terms of flood extent and depth when using GDEMs for flood simulation. However, in contrast to the results of Xu et al. (2021), our findings reveal that utilizing GDEMs does not necessarily lead to an overestimation of maximum flood depths; in fact, it can even result in an underestimation of these depths, attributed to the inherent differences in topography of each area.

An accurate flood risk assessment directly depends on the precision of the flood simulation, specifically regarding the flood extent and other flood features (McClean et al. 2020). This dependency arises because the classification of flood hazards and the calculation of flood damages are conducted as a hierarchical process subsequent to the flood simulation. Any inaccuracies in flood extent and depth, resulting in using various GDEMs, can lead to either overestimation or underestimation of the flood risk assessment. Our results underscore the imperative of selecting an appropriate DEM for flood studies. Notably, reducing the spatial resolution from 1m to 30 m can significantly alter hazard levels. Specifically, this reduction elevates the perceived safety level to H1, which is deemed generally safe for vehicles, people, and structures according to the hazard mapping methodology employed in this study. This leads to an underestimation of higher (H3-H6) hazard levels and an overestimation of lower (H1) hazard levels, both by approximately 50%, when compared to the RM1. Furthermore, the lower the spatial resolution of GDEMs, the greater the number of buildings that are subject to flooding, thereby increasing the vulnerability of these structures. Accordingly, when a model underestimates depth but overestimates extent, it can cause a considerable discrepancy in damage estimations, as observed with a 161% overestimation in NASADEM-30 compared to the RM1, which estimated the damages at $8.6 million. It is worth noting that utilizing GDEMs not only overestimates the flood damage but also alters the spatial distribution of flood damage compared to that estimated by a high-resolution DEM.

Furthermore, the results indicated that the Res_5 m map is a viable alternative to higher-resolution maps for urban flood simulations. This balance between precision and computational efficiency was evident as the shift from a 1-m to a 5-m resolution led to only a modest 6% decrease in the flood zone area, along with a significant 73% overlap in flood zones compared to the RM1. The RMSE index of approximately 0.3 m further confirmed the Res_5 m map's adequacy in accurately classifying flood depths. It also proved efficient in creating hazard maps, which showed minimal variation in hazard classification, and in estimating flood damage, with an underestimation of about 25%. This finding highlights the feasibility of using moderately high-resolution DEMs in urban areas, where they can support effective flood risk assessment and planning without the need for more resource-intensive high-resolution data.

5 Conclusion

Flood modeling is one of the most important criteria for decision-making in flood management, with its accuracy completely dependent on the precision of the topographic maps used as inputs for hydraulic models. This study aims to assess the effectiveness and accuracy of various GDEMs, including ALOS-12.5, ALOS-30, SRTM-30, SRTM-90, and NASADEM-30, specifically in urban flood modeling and risk assessment. To ensure a controlled comparison, the evaluation involves hydraulic modeling for a 50-year return period flood in a densely populated area, using the MIKE FLOOD model, by setting the same simulation conditions but only changing the GDEMs for flooding simulation. These GDEMs were compared with the high-quality Pleiades 1A 1m DEM, serving as the reference.

The results showed significant differences, such as the difference in inundation extents, spatial depth distribution, hazard maps, and flood damages between the GDEMs and the RM1. NASADEM-30 and SRTM-30, despite overestimating flood extents, outperformed others in the spatial distribution of flood extent. This implies that low-resolution DEMs do not necessarily yield more precise results in flood extent—evidenced by the

underperformance of ALOS-12.5. Additionally, utilizing GDEMs in flood simulation generally leads to underestimation of flood depth, resulting in more uniform spatial depth distribution in compared to the RM1. Therefore, flood risk assessments for cities, when conducted using GDEMs, should be interpreted with caution, as there is a likelihood of overestimating the flood extent and underestimating the flood depth. This can lead to inaccurate flood hazard mapping and the overprediction of flood damages.

The inherent inaccuracies of GDEMs, particularly in capturing flood depth and extent, pose significant challenges for urban planning. The dual issues of overestimating flood extent and underestimating flood depth can lead to both overly cautious and insufficiently prepared responses, potentially resulting in the misallocation of resources and inadequate emergency planning. Furthermore, in urban settings, where minor elevation changes can significantly affect flood dynamics, such inaccuracies have profound implications for resource distribution, policy-making, and the effectiveness of flood mitigation strategies.

All in all, while GDEMs have advanced the global execution of flood risk assessments by offering a standardized methodology that facilitates comparisons across different regions, they are not without limitations. Their performance in flood inundation mapping varies, primarily due to differences in data acquisition, processing methodologies, spatial resolution, and inherent errors, which can lead to inaccuracies in flood risk management decisions. Such shortcomings underscore the need for localized studies and caution against considering GDEMs as outright substitutes for more precise DEMs, as recommended by earlier research (Sampson et al. 2016; Schumann and Bates 2018; McClean et al. 2020). Consequently, employing existing GDEMs in conjunction with each other within flood risk frameworks is suggested to comprehensively capture potential uncertainties and their impacts. Despite these drawbacks, GDEMs (in this study SRTM-30 and NASADEM-30) can still serve as a valuable alternative source of data during emergencies, offering crucial insights into the potential extent of urban flooding. This capability underscores the importance of GDEMs as a pivotal tool in immediate crisis management, facilitating the rapid development and implementation of emergency response measures.

To complement the findings of this research, it is recommended to enhance the accuracy and performance of GDEMs in urban flood modeling. This enhancement can be notably achieved by integrating remote sensing data, ground data, and artificial intelligence (AI), especially in regions where data are limited. Such improvements are essential for advancing the reliability of flood risk assessments and management tools.

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Consent to Participate Not applicable.

Consent to Publication The authors give their full consent for the publication of this manuscript.

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